these features lists from a consumer electronics product and a dress are very different!

Let’s focus on the latter: given an image of a dress, find other products in your catalog that look similar: this is a very stimulating problem, even today!

The way we extract features from images completely changed in the last 10 years: in 2009 we used to extract edges, corners and other geometrical features from the images, using dozens of algorithms specialized on the single feature, and then build higher level features by hand (quite literally).

Today, instead, we would use deep learning for the task, training a CNN[1] on a larger dataset and then applying it to all the images in our catalog to generate their feature vector.

Once we have these feature vectors, though, the same question arises, now as then: how do we efficiently search the most similar vectors to a given one?

This is exactly the same problem we illustrated in chapter 8 for 2-D data, applied to a huge dataset (with tens of thousands of images/feature vectors) and where tuples have hundreds of features.